

Kanwaljot Singh

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EDUCATION

- **Cluster Innovation Center, University of Delhi** New Delhi, India
Bachelor of Technology in Information Technology Nov 2022 – Jul 2026
and Mathematical Innovations
CGPA: 8.28/10
Coursework: Computational Analysis of OMICS data, Genes to Genomes, Genomics and Proteomics, In Silico Biology, Data Science, Linear Algebra, Data Structures, Operating Systems, Probability and Statistics, Database Management Systems, Linear Programming and Game Theory

SKILLS

- **Programming:** Python, C++, JavaScript
- **ML/DL Frameworks:** scikit-learn, TensorFlow, PyTorch
- **Data Analysis Tools:** pandas, NumPy, MATLAB, Origin
- **Web/Cloud:** React, Firebase (Auth, RTDB/Firestore, Hosting)
- **Databases:** MySQL
- **Tools:** Jupyter, Arduino IDE
- **Lab/Instrumentation:** Micropipetting, Centrifugation, pH Adjustment, Filtration, Sample Handling

RESEARCH EXPERIENCE

Graphene quantum dots as optical-electrochemical dual mode-biosensors for the detection of dopamine and serotonin biomarkers of depression **DOI: 10.1016/j.jddst.2026.108151**
Prof. Mahima Kaushik, University of Delhi Sep 2024 – Jan 2025

- Synthesized graphene oxide quantum dots (GO QDs) via eco-friendly citric acid pyrolysis.
- Characterized using UV-Vis, FTIR, XRD, DLS, and zeta potential analysis to validate nanoscale dimensions and surface properties.
- Investigated interactions with biomarkers (dopamine and serotonin) for depression-related biosensing.
- Calculated binding constants demonstrating selective dual-analyte detection capability.

AI-Driven Protein–Peptide Interaction Analysis Pipeline Under Internal Review
Dr. J.S. Purohit, University of Delhi Jun 2025 - Sep 2025

- Developed an end-to-end computational pipeline for peptide generation, structural prediction, and protein–peptide docking analysis.
- Leveraged deep learning-based sequence models and structural bioinformatics tools for interaction modeling and validation.
- Performed stereochemical assessment and evaluated binding interactions using established structural and physicochemical metrics.
- Applied the workflow to antimicrobial peptide design and identification of stable peptide–protein binding candidates.

PROJECTS

- **Pathogenicity Prediction of In-Frame Indels Using Protein Language Models** Ongoing
 - Created a deep-learning-based sequence model for in-frame indels using ESM-derived likelihoods to compute Δ scores between wild-type and mutant proteins.
 - Engineered a feature pipeline integrating sequence likelihoods, conservation measures, and physicochemical descriptors for pathogenicity prediction.
- **Remote Photoplethysmography (rPPG) for Contactless Heart Rate Estimation** GitHub

- Developed a contactless heart-rate estimation pipeline using facial video analysis, color-channel extraction, and temporal signal filtering.
- Implemented classical rPPG methods (Green channel, CHROM, POS) for robust pulse signal extraction from facial regions.
- Designed a lightweight deep learning model for heart-rate prediction from spatiotemporal video data.
- Calibrated for Indian skin tones and evaluated performance on short-duration webcam recordings.
- **Hybrid Stacking Model for Corporate Bankruptcy Prediction Using Ensemble Learning GitHub**
 - Developed a machine learning pipeline to predict corporate bankruptcy using financial ratio datasets, addressing class imbalance through SMOTE-based resampling.
 - Designed a hybrid stacking architecture combining XGBoost and neural networks with Logistic Regression as a meta-learner for improved generalization.
 - Performed feature selection and dimensionality reduction using autoencoders and statistical methods to enhance model robustness.
 - Achieved $\sim 99\%$ accuracy with high ROC-AUC (~ 0.995), demonstrating strong predictive performance on benchmark datasets.

CONFERENCE PRESENTATIONS

- Presented at **National Conference on Advanced Materials for Sensors in Jawaharlal Nehru University, New Delhi** **Poster**
 - **Green Synthesis of Graphene Oxide Quantum Dots: Biosensor for Dopamine Detection**
Delivered a poster presentation on the synthesis, characterization, and application of graphene oxide quantum dots (GO QDs) for dopamine detection. Highlighted the eco-friendly synthesis of GO via citric acid pyrolysis and its characterization using UV-Vis spectroscopy, FTIR, XRD, DLS, and zeta potential analysis. Discussed its interaction with dopamine using UV titration, fluorescence studies, and cyclic voltammetry, showcasing its potential in biosensing and nanomedicine applications.